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ARTICLE



Financial Outcomes of Interventions Designed to Improve Retirement Savings: A Systematic Review

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ABSTRACT

This systematic review focuses on examining the effects of interventions that facilitate retirement savings through a financial capability approach, which combines financial education and financial access. Systematic review procedures were used to search for published and unpublished experimental studies in multiple databases and gray literature sources that met eligibility criteria. Four research projects published through May 2020 were eligible for the review. Results suggest that, thus far, there is no clear rigorous evidence that the interventions that use a financial capability approach to promote retirement savings improve individual financial behaviors or financial outcomes. Policy and practice implications are discussed.

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Recent trends indicate that a stronger societal emphasis on retirement savings is warranted (Lusardi et al., 2018). Many Americans are struggling to save for retirement, and may not be on track to have adequate saving for retirement. Even Americans close to retirement age commonly worry about their retirement savings (Board of Governors of the Federal Reserve, 2019). In response to this growing need, interventions are being designed and implemented to improve retirement savings toward the goal of financial stability and well-being in retirement. One promising approach to help people improve retirement savings are interventions designed to improve financial capability. Grounded in the financial capability framework, interventions to promote financial capability pair financial education and a facilitation of access to a mainstream financial product or service related to retirement savings (e.g., employer-provided retirement account, IRA, or Roth IRA). Financial capability interventions are also used to promote long-term financial well-being in earlier parts of the human life cycle, such as financial asset development for lower-income-employed adult populations and children (Grinstein-Weiss et al., 2015; Huang et al., 2014), credit building (Burke et al., 2019) and short-term savings (Grinstein-Weiss et al., 2017). Although varied in terms of populations served, financial education content provided, financial education delivery methods, and financial products or service provided, all

financial capability interventions provide financial education and a particular product or service that can aid wealth creation. Financial capability interventions help a consumer's "... ability to apply financial knowledge and perform desirable financial behaviors to achieve financial well-being" (Xiao & O'Neill, 2016, p. 712).

This study's aim is to conduct a systematic review and meta-analysis to summarize the results of available research on financial outcomes of financial capability interventions designed to improve retirement savings. To the authors' knowledge, this is the first systematic review of financial capability interventions for retirement savings. Systematic reviews offer the highest level of evidence on interventions by summarizing the results of available evidence, and provide guidance about "what works" to practitioners and policymakers.

Background

Policymakers, practitioners, and researchers are focused on the need for U.S. consumers to be better financially prepared for retirement (Board of Governors of the Federal Reserve, 2019; Morrissey, 2019). Due to a political shift away from economic safety nets for families, such as pensions, to one in which families accept more responsibility and risk for their financial wellbeing, families are experiencing greater financial insecurity, families are facing their retirement years with less confidence. The workplace and public/private benefits guarantee far less financial security than they once did (Hacker, 2006). Thus, working-age adults are in increased need for personal saving for their later years. Most families, even those approaching retirement, have little or no retirement savings (Morrissey, 2019). During the prime years for saving for retirement, the percentage of families saving for retirement has decreased from 60% in 2001 to 54% in 2016 (Morrissey, 2019). Adults close to retirement hold more debt and are less financially secure than in previous years (Lusardi et al., 2018).

Today, only 22% of non-retirees have a defined benefit pension, which is income that is guaranteed for life. Among non-retirees with savings, 54% save in a defined contribution plan (i.e., 401(k) or 403(b)) (Board of Governors of the Federal Reserve, 2019). These accounts put all of the major risks and responsibilities, such as the risk that investments will decrease in value, the risk of outliving one's retirement savings, the risk of making decisions about investments, on the non-retiree (Hacker, 2006). Others have savings within other types of accounts, such as IRAs, real estate, business, and other (Board of Governors of the Federal Reserve, 2019). The historic shift from pensions to account-type savings plan has increased the financial insecurity for the majority of the U.S. population. The populations most negatively affected include consumers who are lower income, black, hispanic, non-college-educated, single, and women (Morrissey, 2019). Although only designed to be supplemental, Social Security retirement benefits will likely be the main source of retirement income for these populations (Social Security Administration, 2020).

The issue of inadequate financial preparation for retirement is particularly relevant for social work and other human service practitioners who engage with working families and older adults and their families. They will be working with those whose lack of retirement savings and financial supports will also place a heavier burden the nation's social safety net in the coming decades (Miller et al., 2017). This emerging retirement crisis shed's a spotlight on the need to design and implement interventions to increase retirement savings to assist younger populations to increase retirement savings.

Financial outcomes of interventions

Financial education is a commonly used method to promote or facilitate retirement savings (e.g., Atkinson et al., 2015; Boyer et al., 2019). Individual and systematic studies on interventions that only offer financial education or advice to promote retirement savings thus far have demonstrated mixed financial outcomes. For example, Atkinson, et al. (2015) conducted a review and found that some financial education interventions aimed at increasing long-term saving and investment had promising results, while others have mixed outcomes. In their meta-analysis, Miller et al. (2014) found that retirement-focused financial education had weak effects on retirement savings. Similarly, Hung and Yoong (2010) found that unsolicited financial advice has no impact on savings and investment decisions. Beshears et al. (2015) found that the presence of peer information actually decreased the savings of non-participants who were ineligible for 401(k) automatic enrollment. Choi et al. (2004) found that the design of a 401(k) plan impacts participation in the plan, the distribution of employee contribution rates and other aspects. While education is a relatively low-cost intervention, the impacts of education-only interventions to promote retirement savings have not yet demonstrated needed impact on retirement savings. As a result, other types of interventions aimed at increasing retirement savings are being developed and tested.

Financial capability conceptual framework

In recent years, the term "financial capability" is often considered synonymous with financial knowledge, skills, and behavior (Sherraden, 2013). However, to be financially capable, people need both financial knowledge and skills, and access to financial products and services that enable them to act in their own best interest (Sherraden, 2013). In the past, financial education has been a central focus of efforts to shape people's financial behavior with increased knowledge and skills. However, previous individual and systematic review studies on financial education have demonstrated that financial knowledge alone is insufficient to show large effects on financial behaviors and financial outcomes (Fernandes et al., 2014; Hastings et al.,

2013; Kaiser & Menkhoff, 2016; Miller et al., 2014; Mitchell & Lusardi, 2015). These previous studies have not been guided by the financial capabilities framework that combines financial education and access to retirement savings accounts.

The financial capability framework is grounded in the idea of capabilities of philosophers Sen (1999) and Nussbaum (2007). They propose that capability captures people's real possibilities in life for development, considering both their internal capabilities (e.g., knowledge, skills, and abilities) and the conditions and opportunities in their environment (e.g., access to resources, products, and services). Internal capabilities and external environment interact with each other to provide potential for development. Building on the capabilities theoretical work, Sherraden (2013) theorized that financial capability requires a combination of financial knowledge and skills, and access to appropriate financial products and services. Due to this interaction, she maintains that it is an "expression of an individual's economic and social position and influence in relation to mainstream financial and economic institutions" (p. 4). Financial options that are within the realm of possibilities shape people's vision of possibilities and thus influence behavior. Evidence suggests that the combination of the two results in financial behaviors and financial access that facilitate financial well-being (Theodos et al., 2015; Collins & Urban, 2016; Huang et al., 2013; Curley & Robertson, 2017). Some retirement savings interventions provide both financial education to increase participant knowledge and skills, and access to a retirement account (e.g., Collins & Urban, 2016; Goda et al., 2012). The interaction of the knowledge and skills with access facilitates individuals' ability to *apply* their knowledge and skills through their financial behaviors, such as opening an account and saving money for retirement purposes, increasing their rate of savings, or more knowledgeably investing their retirement savings.

Purpose of the present study

The purpose of this systematic review was to inform practice and policy by examining and synthesizing evidence of the effects of interventions designed to improve financial capability through financial education and retirement savings accounts. We aimed to answer these research questions: (a) What are the effects on financial behavior and financial outcomes of interventions designed to improve financial capability through retirement savings accounts? and (b) Does study intervention (dosage, duration, type) or sample (age) characteristics relate to the magnitude of effect size?

Methods

Systematic review methodology was utilized for all aspects of the search, selection, and coding of studies. Meta-analysis was planned to quantitatively synthesize effects of interventions. Campbell Collaboration procedures and guidelines for systematic review and meta-analytic methods were used (Campbell Collaboration, 2020). The protocol and data extraction form for this review have been published (Birkenmaier et al., 2019). This project was part of a larger systematic review project examining financial capability interventions across age groups and intervention types.

Search strategy

The comprehensive search included multiple electronic databases, gray literature sources (see Table 1), websites and reference lists of reviews and relevant studies (Kugley et al., 2017). To search the electronic databases, the search terms used in combinations were: (a) intervention: (financial OR economic OR bank) AND (education OR knowledge OR literacy), AND (capability OR access OR inclusion OR exclusion OR attachment) and (b) report type: (evaluation OR intervention OR treatment OR outcome OR program OR trial OR experiment OR “control group” OR “controlled trial” OR quasi-experiment” OR random*). The authors searched for studies from the databases’ earliest date through May 2017. An updated search was conducted for

Table 1. Electronic Databases and Sources Searched.

Name of Database	
ABI/INFORM	Education Source
Academic Search Complete	JSTOR
Bloomberg Professional Service	PAIS Index
Business Source Premier	PsychInfo
Database of Research on International Education	Public Affairs Index
Dissertation & Theses Global	Social Science Research Index
EconLit	Social Sciences Citation Index
ERIC	Social Work Abstracts
	Sociological Abstracts
Research Registries	
The National Institute for Health U.S. National Library of Medicine (Clinicaltrials.gov)	
The International Clinical Trials Registry Platform	
Government Websites	
Consumer Financial Protection Bureau	
“Finlitedu.org” (World Bank and the OECD Global Partnership for Financial Inclusion)	
Federal Reserve Banks (12)	
U.S. Treasury Department	
Organization Websites	
Alliance for Financial Inclusion (AFI)	Global Partnership for Financial Inclusion
Center for Social Development	Micro Financing Opportunities
CGAP	Russell Sage Foundation
Citi Foundation	Urban Institute
Employee Benefit Research Institute	

studies from May 2017 – May 2020. The updated search was conducted using the same process as the original search.

One reviewer conducted the initial search of all sources and saved the search results in an electronic format. After the initial search, using EndNote (sold by Clarivate Analytics, 2020), the reviewer examined titles and abstracts and discarded results that were obviously ineligible (e.g., non-empirical reports, etc.). For those possibly eligible, the reviewer retrieved the full reports. Using the Covidence dashboard (sold by Clarivate Analytics, 2020), two reviewers then independently screened each of the reports for eligibility using a screening instrument (available from the authors). Coding was compared, and discrepancies were resolved through consensus by meeting together to review any disagreements and checking the source documents.

Eligibility criteria

To be eligible, the study must have been examining the effects of a financial capability intervention, which is defined as including a financial education component and access to a retirement account. To meet the criteria for delivering financial education for this review, interventions must have delivered information about: (1) a variety of general financial concepts and behaviors (such as using a formal education curriculum that covers the time value of money); (2) a specific financial topic (such as a formal or informal information about savings); and/or (3) a specific product (such as a retirement account). Financial education could have been high-intensity, such as one-on-one delivery, or low intensity, such as flyers, e-mails, texts, videos, or online delivery.

To meet the criteria for access to a financial product or service, interventions must have facilitated access to a retirement account by automatic enrollment or facilitating access to such accounts, or aimed at increasing retirement account contributions. For study designs, studies must have used a prospective randomized controlled trial (RCT) or quasi-experimental (QED) research design. Studies must have used a comparison group that received an alternative intervention, a placebo condition, business as usual, or no intervention. Because we were primarily interested in the financial outcomes, studies must have reported at least one of the following outcomes: behavior change or financial outcome. *Behavior change* refers to changes in the financial behaviors of participants, such as opening an account. *Financial outcome* refer to implications of behavior change, such as higher fund balance in a retirement account. Outcome measurement could have been conducted using standardized or unstandardized instruments, and self- or other-reported or researcher administered measures. When authors reported additional outcomes, we noted these, but other outcomes were not used in determining eligibility. Reviewers planned to pool effects based on the type of measure used (e.g., observational measures pooled with

observational measures). To be included in the meta-analysis, study authors must have reported enough information to calculate an effect size. If sufficient information to calculate an effect size was not provided, we requested the necessary information from study authors. Studies with participants of all ages and in all settings were included. For similarity of economic contexts, studies must have been conducted with participants in any of the 35-member countries of the Organization for Economic Co-Operation and Development (OECD). Countries that participate in OECD tend to be wealthier countries and have more similar economic and financial systems. This review includes studies that were published through May 2020. The reviewers included measurement points at posttest and all follow-up time points. We planned to synthesize studies that reported similar follow-up time points (i.e., up to 3 months, 3–6 months, etc.) if more than two studies reported sufficient data.

Two authors independently assessed risk of bias in all included studies using the Cochrane Collaboration's risk of bias tool (Higgins et al., 2011). The authors assessed risk of bias for each of the seven following domains as “low”, “high”, or “unclear”: sequence generation, allocation concealment, blinding of participants and personnel for all outcomes, blinding of outcome assessors for all outcomes, incomplete outcome data for all outcomes, selective outcome reporting, and other potential sources of bias (i.e., researcher allegiance, funding source). Following independent coding by two review authors, they met to identify and resolve any discrepancies through consensus. No studies were excluded due to bias.

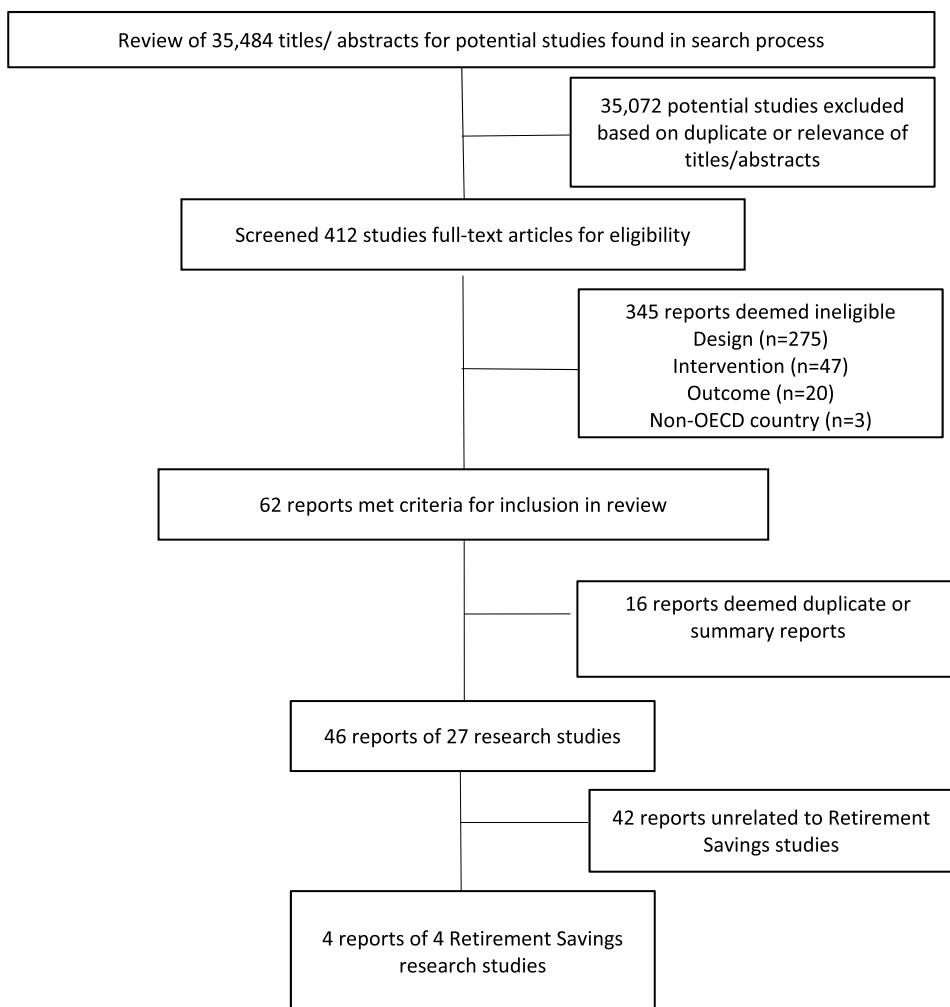
Synthesis procedures and statistical analysis

The review authors conducted descriptive analyses on variables of interest from all included studies to provide information regarding: (1) publication type; (2) study design; (3) sample size; (4) target beneficiaries; (5) race; (6) gender; (7) income; (8) intervention target; (9) intervention goal; (10) financial capability intervention type; (11) financial education; and (12) age of beneficiaries. Unfortunately, there were not a sufficient number of studies that measured the same types of outcomes at similar time points or reported sufficient data to calculate an effect size, thus, a meta-analysis was not possible. We calculated Hedges' *g* effect sizes when sufficient information was available and reported effect sizes as well as results reported by the primary study authors. Some studies were published in multiple reports; therefore, we were careful to ensure that each unique study was counted only once. Multiple reports from the same study were included as secondary reports to the (parent) study. Since meta-analysis was not possible, we reported all measures and time points for each outcome of interest for each study narratively in the results section and in tables.

Results

As seen in [Figure 1](#), electronic searches of bibliographic databases and other sources identified a total of 35,484 results. Titles and abstracts were screened for relevance and 35,072 were excluded as duplicates or deemed inappropriate.

Study Screening Process



Note: The original search was conducted from through May, 2017. A second round of searching was conducted from May, 2017 through May, 2020.

Figure 1. Study Screening Process.

Note: The original search was conducted from through May, 2017. A second round of searching was conducted from May, 2017 through May, 2020

The full text of the remaining 412 potential studies was reviewed and screened for eligibility by two independent coders. We excluded 345 reports that were deemed ineligible (list available from the first author) and included 62 reports that met inclusion criteria. Of the 62 reports, 16 were deemed duplicates or summary reports. Of the remaining 46 reports, 27 were unique studies (i.e., used unique samples) that were included in the larger project. Four of these were unique retirement account studies that met inclusion for this review.

Study characteristics

Four studies met the inclusion criteria (See [Table 2](#) includes study characteristics). In the four studies, data were collected on financial behavior and/or financial outcomes of the study participants using unstandardized instruments, and included pre- and posttest and administrative data. Two studies were randomized control trials (RCTs) and 2 were quasi-experiment designs (QEDs). One study has a sample size in the range of 1–999, one has a sample size in the range of 1,000–2,999, and two studies had samples in the range of 13,000–16,999. Employees ($n = 3$) and tax filers ($n = 1$) were the two targeted populations. There were three different intervention activities: new employee retirement signup ($n = 1$), (current) employee account opening/contribution changes focused on current employees ($n = 2$); and a matched contribution to retirement account with a portion of the tax refund ($n = 1$). The mean age of adult participants was unknown in three studies and 45 years old in one study. Three studies included participants from all income levels, and one included only low-income participants. Only one study reported predominant race (White). No studies assessed previous participant financial education. In one study, 44% of the participants were male, and in one study, 19% of the participants were male. The remaining two studies did not report sample gender statistics. For intervention characteristics, the majority of the studies provided low-intensive financial education ($n = 3$). The intervention duration ranged from one to seven months.

Financial outcomes

Across the four studies, three measured account opening: Collins and Urban (2016), Duflo et al. (2006), and Lusardi et al. (2009). Account opening was measured at the following time points: (a) end of treatment (Duflo et al., 2006); (b) end of treatment, six and 12 months later (Collins & Urban, 2016), and (c) within 30 days after treatment (Lusardi et al., 2009). Two studies measured retirement savings rates: Collins and Urban (2016); and Goda et al. (2012). Retirement savings rate was measured at the following time points: (a) end of treatment, six and 12 months later (Collins & Urban, 2016); and (b) end of treatment (Goda et al., 2012). One study also measured budgeting (Collins and

Table 2. Characteristics of Included Studies.

Study	Study Design	Intervention Population	Sample Size	Intervention Length	Intervention Method	Intervention Content	Outcomes Type	Data Source	Results
Collins and Urban (2016)	QED (Author states RCT)	Credit union employees	n = 1,001 Tx (n = 301) Comparison (n = 700)	10 module online financial education	Current employee retirement account opening or contribution change	Financial education and retirement plan enrollment or contribution change	Account Opening Retirement Savings Amount Retirement Savings Rate Budget Savings Amount	Pre-post survey; administrative data	Study authors reported increases of 6% for account opening, almost 6% for rate of budget creation, 3.7–3.8%, for saving participation, 40.4% for retirement savings rate, and 3.8% for emergency savings.
Duflo et al. (2006)	RCT	Tax filers	n = 13,952 Tx Group 1 n = 4,521 (20% match) Tx Group 2 n = 4,722 Control group n = 4,719	1 session	Matched contribution to retirement account with portion of tax refund	Financial education and matched retirement fund contribution Tx Group 1–20% match Tx Group 2–50% match	Account Opening Savings Amount	Administrative Data	Tx Group 1 Account opening (OR = 2.79; CI = 2.28–3.41) Tx Group 2 Account opening OR = 4.67; CI = 3.87–5.64 Saving Amount data unreported

(Continued)



Table 2. (Continued).

Study	Study Design	Intervention Population	Sample Size	Intervention Length	Intervention Method	Intervention Content	Outcomes Type	Data Source	Results
Goda et al. (2012)	RCT	University employees	$n = 16,881$	Seven months	Current employee retirement account opening or contribution change	Varying amounts of low-intensity financial education (direct-mail) and supported sign-up for retirement accounts	Retirement Savings Amount Retirement Savings Rate	Administrative data	Retirement Saving Amount Tx group 1 $d = -0.014$; 95% CI = $-0.06-0.03$; tx group 2, $d = -0.02$; 95% CI = $-0.02-0.07$; tx group 3, $d = -0.001$; 95% CI = $-0.04-0.04$ Retirement Saving Rate Tx group 1, $d = -0.02$; 95% CI = $-0.06-0.02$; tx group 2, $d = 0.0236$, 95% CI = $-0.02-0.07$; tx group 3, $d = 0.03$; 95% CI = $-0.01-0.07$
			Control group ($n = 4,063$) Tx group 1 $n = 4,434$, Tx group 2 $n = 4,353$, Tx group 3 $n = 4131$.						
Lusardi et al. (2009)	QED	New female and low-income University employees	$n = 124$ (treatment group $n = 64$, comparison group $n = 60$).	1 session	New employee retirement account sign-up	Flyer inserted in the employee benefits packet, a visual presentation, and an instructional video) and supported access to retirement accounts.	Account Opening	Posttest survey data	Study authors reported a 56.2% increase in account opening within 30 days of the intervention compared to the comparison group.

Table 3. Bias Across Studies.

Bias domains	Duflo et al., 2006	Collins & Urban, 2016	Lusardi et al., 2009	Goda et al., 2012
Sequence generation	Low	High	High	Low
Incomplete outcome data	Low	Unclear	Unclear	Unclear
Blinding of outcome assessors	Low	Low	Low	Low
Allocation concealment	Unclear	Unclear	Unclear	Low
Blinding of participants and personnel,	Unclear	Unclear	Unclear	Unclear
Selective outcome reporting	Unclear	Unclear	Unclear	Unclear
Other (Research Allegiance)	High	High	High	Unclear

Urban (2016)). Budgeting was measured at the end of treatment, and six and 12 months.

The financial outcome of saving amount was measured in three of the four studies at: (a) the end of treatment (Duflo et al., 2006; Goda et al., 2012); and (b) monthly between four and 11 months after treatment (Collins & Urban, 2016). Next, we report effect sizes of outcomes when enough information was reported to calculate, or author-reported outcomes if we could not calculate effect sizes due to missing data.

Retirement account studies

Duflo et al. (2006)

This study used a randomized control trial to examine the effects of a retirement account intervention with tax filers in St Louis, Missouri (U.S.) who used a private paid tax preparation company. Participants were all tax filers of any income using the service during the one-month intervention period during tax season, and were individually randomly assigned to the treatment or control group. Tax preparers delivered low-intensity financial education (brief, face-to-face, only related to their taxes and retirement savings), along with matches of 20% (treatment group 1) and 50% (treatment group 2) of participant contributions into the self-retirement funds (IRAs) from the amount of money refunded to them through their tax returns. The control group received no financial education and zero match. The total sample was $n = 13,952$ (treatment group 1 $n = 4,521$ (20% match), treatment group 2 $n = 4,722$ (50% match), and control group $n = 4,719$ (zero match and no financial education)). Using administrative data, study authors analyzed savings amounts and account opening at the end of treatment. Study authors did not provide sufficient information to calculate an effect size for savings amount (i.e., missing standard deviation for post-intervention treatment and control groups). The study author reported effects of the intervention on account opening for the treatment group 1 that received a 20% match compared to the control group was large ($OR = 2.79$; $CI = 2.28-3.41$). For treatment group 2 that received a 50% match compared to the control group, the effect of the intervention was also

large (OR = 4.67; CI = 3.87–5.64). As seen in [Table 3](#), we assessed the risk of bias to be low for bias domains of sequence generation, incomplete outcome data, and blinding of outcome assessors. The risk of bias was assessed to be unclear for allocation concealment, blinding of participants and personnel, and selective outcome reporting. The risk of bias was assessed to be high for other (research allegiance, as the study authors designed the intervention).

Collins and Urban (2016)

This study employed a quasi-experimental design to examine the effects of an intervention that paired financial education with supported signup or changes to an employer-provided retirement plan for the employee participants ($n = 1,001$) of 45 Wisconsin credit unions. Treatment participants completed a 10-module online financial education program (high-intensity) and could either sign-up for a retirement plan or continue or alter their retirement savings behavior. The length of the intervention was about four months. Participants were assigned to condition based on their credit union. Comparison group members received nothing. Using pretest and posttest survey data and administrative data before and after treatment was offered, study authors examined the financial behavior outcomes of account opening, retirement savings rate, budgeting, as well as the financial outcome of savings (emergency) amount. The total sample was $n = 1,001$ (treatment group $n = 301$, comparison group $n = 700$). Authors did not report sufficient data to calculate effect sizes (i.e., treatment and control group baseline and post-intervention means and standard deviations, treatment and control group percent successful and unsuccessful for account openings, etc.). The study authors reported that the intervention increased account opening by 6%, and the rate of creation of a budget by almost 6% more than the comparison group. The authors also reported that the intervention increased the retirement savings participation by 3.7–3.8%, the retirement savings rate by 40.4%, and emergency savings by 3.8% as compared to the comparison group. As seen in [Table 3](#), the risk of bias was judged as high for this study for sequence generation and research allegiance – the study authors designed the intervention. The risk is unclear for allocation concealment, blinding of participants and personnel, incomplete outcome data, and selective outcome reporting. The risk of bias is low for blinding of outcome assessors.

Lusardi et al. (2009)

This study used a quasi-experimental design to examine the effects of a retirement-focused intervention geared toward female and low-income employees of Dartmouth College, populations that traditionally save little for retirement. They had a total sample of $n = 124$ (treatment group $n = 64$, comparison group $n = 60$). The study authors measured outcomes at the end of the intervention period, which was six months. The treatment was delivered

during new employee orientation, the length of which is not reported. The study authors scheduled the intervention during the first six months of the year within which females and low-income employees are more likely to attend new employee orientation, as compared to higher-income faculty members (July–December). During the one-time new employee orientation, the treatment group received low-intensity financial education (flyer inserted in the employee benefits packet, a visual presentation, and an instructional video) and supported access to retirement accounts. The comparison group consisted of employees who had completed new employee orientation a year prior to the intervention during the same season, when they had not received the low-intensity financial education, but could have signed up for the retirement plan. Using posttest survey data, study authors studied the outcome of account opening. The study authors did not include data needed to calculate an effect size for account opening (i.e., percent of treatment and control groups that successfully opened an account and did not open an account). Study authors reported that the intervention resulted in a 56.2% increase in account opening within 30 days of the intervention compared to the comparison group. As seen in [Table 3](#), the review authors assessed the risk of bias as high for sequence generation, and research allegiance as the study authors designed the intervention, and unclear for allocation concealment, blinding of participants and personnel, incomplete outcome data, and selective outcome reporting. The risk of bias is low for blinding of outcome assessors.

Goda et al. (2012)

This study used a randomized control trial to examine the effects of an intervention using low-intensity financial education (direct-mail) and supported signup for retirement accounts with employees of the University of Minnesota. The study authors randomly assigned employees by departments ($n = 16,881$) to one control group ($n = 4,063$) and three treatment groups ($n = 12,828$; group 1 $n = 4,434$, group 2 $n = 4,353$, and group 3 $n = 4,131$). The control group received nothing, and the three treatment groups received varying levels of detailed financial education about retirement. The intervention lasted for seven months, and the intervention was evaluated at the end of treatment. Study authors used administrative data to study savings amount and retirement savings rate. The effect of the intervention on the saving amount, compared to the control group, was trivial and not statistically significant for all treatment groups (treatment group 1, $d = -0.014$; 95% CI = -0.06 – 0.03 ; treatment group 2, $d = -0.02$; 95% CI = -0.02 – 0.07 ; and treatment group 3, $d = -0.001$; 95%, CI = -0.04 – 0.04). The effect of the intervention on the retirement saving rate was also trivial and not statistically significant for each of the treatment groups compared to the control group (treatment group 1, $d = -0.02$; 95% CI = -0.06 – 0.02 ; treatment group 2, $d = 0.0236$, 95% CI = -0.02 – 0.07 ; and treatment group 3, $d = 0.03$; 95%,

CI = $-0.01-0.07$). As seen in Table 3, the review authors assessed the risk of bias to be low for the bias domains of sequence generation, allocation concealment and blinding of outcome assessors, and unclear for blinding of participants and personnel, incomplete outcome data, selective outcome reporting and other.

Discussion and Implications

This systematic review examined the effects of retirement savings interventions using a financial capability approach on financial behaviors and financial outcomes. Three of the studies used a low-touch financial education approach (one-time or a small amount of information) (Duflo et al., 2006; Goda et al., 2012; Lusardi et al., 2009), while one used a more intensive approach (Collins & Urban, 2016), and all four provided access to retirement savings vehicles. Even though a meta-analysis could not be conducted, two study authors did identify positive effects on outcomes; one study found large effects on account opening (Duflo et al., 2006); and another found small effects on saving amount and retirement saving rate (Collins & Urban, 2016). Further, two study authors reported positive outcomes of their interventions on account opening (Collins & Urban, 2016; Lusardi et al., 2009), and one found positive outcomes on budgeting, saving amount, and retirement saving rate (Collins & Urban, 2016). In sum, only one study reported a large effect on one behavior change – account opening (Lusardi et al., 2009). The other three studies either had small effect sizes that were statistically insignificant, or we were unable to calculate the effect size for the outcomes. Because quantitative synthesis was not possible, we need to caution against drawing conclusions about the effectiveness of the included interventions or trying to generalize results based on one study or by doing vote counting (i.e., simply adding up the number of studies that found positive or negative outcomes). While this review is the first to examine financial capability interventions aimed at increasing retirement savings, these findings are consistent with findings of studies of interventions that use only financial education or advice to effect retirement savings. Previous studies of these efforts have mostly found mixed, weak, or null effects on related financial behavior and/or financial outcomes (Atkinson et al., 2016; Beshears et al., 2015; Hung & Yoong, 2010; Miller et al., 2014).

Overall, results suggest a lack of clear rigorous evidence of the effects of retirement savings interventions using a financial capability approach on financial behaviors or financial outcomes. While half of the studies used random assignment, the studies also had some important methodological weakness. For example, only one of the studies reported whether they used a manualized intervention or measured study fidelity (Collins & Urban, 2016). Two of the four studies did not report their attrition rate (Duflo et al., 2006; Goda et al., 2012). The assessment of risk of bias in the domains varied from

low to high across the studies. In addition, the studies did not report the length of time that passed between the intervention, that lasted between one and seven months across the studies, and outcome assessment, making it difficult to discern whether any of the outcomes were assessed in the long term.

This systematic review points to the need for more evidence on the financial capability approach to retirement savings interventions. The needs for financial planning for retirement are escalating, especially in the wake of the economic shocks stemming from the COVID-19 pandemic, as well as the growing population of older adults. Social work scholars should expand efforts to meet the need for more evidence about related interventions. Future research on this topic should include high-quality studies with independent samples. Studies should also report standardized outcomes, both short- and long-term, that would enable future meta-analysis to build a body of evidence about this approach to retirement savings interventions in terms of immediate and longer-term effects on financial behavior and outcomes. More definitive evidence about whether the combination of financial education and financial access for retirement savings provides financial benefits to clients on specific outcomes could help build an even stronger constituency to support policy and practice development. Governments, nonprofit and for-profit policy actors, and social workers can promote standardization of outcome measures, measurement time points post-intervention, and other aspects of the service delivery and research methods to build strong evidence about the intervention.

Limitations

This review was guided by Campbell's standards and policies for the conduct and reporting of systematic reviews to ensure a rigorous and transparent review designed to minimize bias and error. We made every attempt to search for published and unpublished studies, but only published studies met inclusion criteria, thus the results may be positively skewed. Although we conducted a rigorous search process, we may have missed relevant studies in our search. The included studies presented with various risks of bias and could not be quantitatively synthesized, thus caution must be used when interpreting findings.

Conclusion

Additional work is needed to design interventions to increase retirement savings. In the absence of significant reform to Social Security, pensions, and other income sources, the importance of private savings will continue to increase. Social workers and other human service practitioners can continue to play a significant role in educating decision-makers and advocating for shoring up current public and private retirement saving programs (Kingson

et al., 2014) and increasing the number of interventions. Most importantly, more conclusive evidence about interventions is needed to guide policymaking toward effective programming to avoid a looming crisis of adults financially unprepared for retirement.

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